

Hope Donglo, PhD

Lecturer, Department of Physics | University of Ghana, Legon

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Professional Profile

Theoretical nuclear physicist (PhD, Université de Caen Normandy; Marie Skłodowska-Curie – COFUND Fellow) specialising in nuclear reaction dynamics and superheavy element synthesis, with research experience at CEA/GANIL, INFN Legnaro, and the Universidad Autónoma de Madrid. Develops theoretical models of heavy-ion reactions and superheavy element synthesis, employing stochastic methods, uncertainty quantification, and scientific computing to investigate reaction mechanisms and improve predictive nuclear theory. Contributes to development and validation of the KEWPIE3 code.

Rank:	Lecturer
Highest Qualification:	PhD, Theoretical Nuclear Physics (Université de Caen Normandy, 2024)
Date of Appointment (UG):	1 August 2026
Date of Assumption of Duty:	17 August 2026
Area of Specialisation:	Nuclear reaction theory; superheavy element synthesis; uncertainty quantification

Education

- 10/2021 – 12/2024 Ph.D. in Theoretical Nuclear Physics**
University of Caen Normandy, France
 - Dissertation:* Study of Reaction Mechanisms for the Synthesis of Superheavy Elements
 - Marie Skłodowska-Curie – COFUND Fellow; research conducted at CEA/GANIL (France)
 - Supervisor:* Dr. Dieter Ackermann (GANIL/CEA)
- 09/2018 – 10/2020 Joint Master's in Nuclear Physics (Erasmus Mundus)**
Universities of Seville, Autonomous Madrid, Padua, and Caen
 - Competitive international scholarship; training across four European universities
 - Thesis:* Nuclear Structure Studies with Symmetry-Conserving Configuration Mixing Method
 - Supervisor:* Dr. Tomás Raúl Rodríguez Frutos
- 09/2012 – 12/2016 Bachelor of Science in Physics**
University of Ghana, Legon — thesis conducted at GAEC/NNRI
 - Thesis:* Evaluation of Safety Status of the Ghana Research Reactor-1 (GHARR-1)
 - Supervisors:* Dr. Allison F. Hughes (UG Physics) and Dr. Joseph K. Gbadago (GAEC/NNRI)

Academic and Research Appointments

- 08/2026 – Lecturer, Department of Physics**
College of Basic and Applied Sciences (CBAS), University of Ghana, Legon
 - Course allocation to be determined
- 10/2021 – 12/2024 Doctoral Researcher, Theoretical Nuclear Physics**
CEA/GANIL, Caen, France
 - Modelled nuclear reaction mechanisms using stochastic simulation methods (Langevin dynamics), covering the full synthesis pathway for super-heavy element production
 - Built and validated HPC simulation pipelines (KEWPIE2/KEWPIE3 framework); predicted production cross-sections for $Z = 119$ and $Z = 120$ nuclei
 - Applied Bayesian optimisation and machine learning methods for nuclear model parameter estimation and uncertainty quantification
 - Performed systematic benchmarking of competing theoretical models with rigorous uncertainty analysis
- 03/2020 – 09/2020 Graduate Research Assistant**
Autonomous University of Madrid (UAM), Spain
 - Simulated 27 argon isotopes ($A = 32\text{--}58$) using the Symmetry-Conserving Configuration Mixing method; mapped structural evolution and shape coexistence across the isotopic chain
- 09/2019 – 01/2020 Graduate Research Assistant**
GANIL, Caen, France
 - Investigated nuclear deformation dynamics using the Geometric Collective Model
 - Identified γ -unstable shape coexistence and performed numerical simulations of quadrupole dynamics

• **03/2019 – 06/2019 Graduate Research Assistant**
National Institute for Nuclear Physics (INFN), Legnaro, Italy

- Optimised radiation detectors (EJ-301, LaBr₃, BaF₂) for neutron/gamma discrimination; achieved 0.01 ns/channel TOF calibration and <5% LaBr₃ energy resolution deviation
- Developed automated data acquisition and analysis pipelines for neutron time-of-flight spectroscopy

Publications

• **Peer-Reviewed**

D. Ackermann, F. Agert, A. Bahini, D. Boilley, **H. Donglo**, S. Kumar, J. Piot, H. Savajols, Ch. Stodel, and the S³ Collaboration (2026). "Probing SHN Structure and Stability – synthesis and decay studies at S³." *Acta Physica Polonica B*, Vol. 19 (2026).

• **Preprint (HAL)**

H. Donglo and T. R. Rodríguez Frutos. "Nuclear Structure of Even-Even Argon Isotopes within the Axial Symmetry Conserving Configuration Mixing Framework." Deposited on HAL (hal-05677054).

• **PhD Thesis**

H. Donglo (2024). *Study of Reaction Mechanisms for the Synthesis of Superheavy Elements*. University of Caen Normandy.

Ongoing Research Projects

• **KEWPIE3 Development and Validation**

Validation and publication of KEWPIE3, the new version of the KEWPIE2 code for fusion-evaporation reaction cross-section predictions in superheavy element synthesis. Contributes to improved compound nucleus formation probability calculations using Langevin-based stochastic approaches.

• **In Preparation**

H. Donglo, F. Agert, D. Boilley, T. Cap, M. Kowal, and D. Ackermann. "Revisiting Fusion Dynamics in the Fusion-by-Diffusion Model for Fusion–Evaporation Production Cross Sections." Targeting *Journal of Physics G*.

• **In Preparation**

H. Donglo, D. Boilley, F. Cap, and D. Ackermann. "Constraining the Compound Nucleus Formation Probability in Superheavy Element Synthesis through Systematic Uncertainty Analysis." Targeting *Physical Review C*.

Research Interests

Nuclear reaction theory (fusion-evaporation, superheavy element synthesis); nuclear structure (collective and mean-field models); Langevin dynamics and stochastic simulation; Bayesian inference and uncertainty quantification for nuclear model parameters; machine learning methods for nuclear data analysis; simulation-assisted nuclear safety analysis; nuclear fission.

Teaching and Supervision

Teaching Experience

2022–2024 BSc Intern Co-Supervisor

GANIL/CEA, Caen, France

Supervised final-year interns on nuclear structure projects; guided Python-based simulation workflows, uncertainty analysis, and scientific writing.

09/2016–08/2017 Undergraduate Teaching Assistant

University of Ghana, Department of Physics, Legon

Supervised laboratory sessions for 150+ undergraduates across experimental physics modules; designed structured data analysis exercises and provided individual mentoring in scientific computing. Supported curriculum delivery alongside academic staff.

2017–2021 Physics and Mathematics Tutor (Intermittent, Part-time)

Greater Accra, Ghana

Delivered secondary-level instruction in physics and mathematics; adapted teaching approaches to diverse learner backgrounds and ability levels.

Teaching Areas

- Introductory Physics and Laboratory Instruction
- Nuclear and Particle Physics
- Quantum Mechanics and Quantum Field Theory
- Mechanics, Thermodynamics, and Electromagnetism
- Computational and Scientific Programming Methods for Physics

Teaching Philosophy

Committed to inquiry-based, student-centred physics instruction that integrates conceptual reasoning, laboratory practice, and computational problem-solving. Championing inclusive, research-informed STEM education.

Fellowships and Awards

- **Marie Skłodowska-Curie Actions – COFUND Fellow** (2021–2024) — European Commission, fully funded doctoral fellowship

- **Erasmus Mundus Scholar** (2018–2020) — European Commission, joint MSc scholarship

Professional Development and Training

10/2025 – 09/2026 Data and ML Engineering Certification (Bac+5, RNCP Level 7) — *OpenClassrooms, France Travail-funded*

Professional certification in scalable data pipelines, MLOps, and machine learning applications, with relevance to computational physics instruction and research methods teaching.

2024 Research Integrity Certification — Université de Bordeaux

Conferences and Scientific Presentations

Seminars and Conference Presentations

- Study of Reaction Mechanisms for the Synthesis of Superheavy Elements. *29th Nuclear Physics Conference*, Poland, September 2023.

Workshop Participation

- GRIT Workshop — Silicon Detectors for Low-Energy Nuclear Physics, GANIL, Caen, France, 2023.
- Workshop on Targets and Ion Sources, GANIL, Caen, France, 2023.
- LISE Workshop — Nuclear Structure and Reactions, GANIL, Caen, France, 2022.
- Physics with SPIRAL2 Heavy Ion Beams, GANIL, Caen, France, October 2022.

Schools and Training

- **CEA NUMERICS Summer School** (2023) — HPC and numerical simulation, CEA
- **IN2P3 School of Statistics** (2022) — Statistical methods for nuclear and particle physics
- **R & D Project Management** (2022) — CEA INSTN
- **Machine Learning and Data Analysis** (2020) — University of Caen

Technical and Computational Skills

- **Nuclear Modelling:** Langevin stochastic dynamics; KEWPIE2/KEWPIE3 framework; cross-section prediction; nuclear structure models (collective, mean-field, shell).
- **Uncertainty Quantification and Statistics:** Bayesian inference and optimisation; uncertainty propagation; model comparison and validation; IN2P3-certified statistical methods for nuclear and particle physics.
- **Machine Learning and Data Analysis:** ML for nuclear parameter estimation; Python scientific computing; SQL; data pipeline development (PostgreSQL, Debezium, Redpanda, Spark/Delta Lake, Superset).
- **Scientific Software Development:** Modular C++/Python for HPC environments; version-controlled workflows (Git); reproducible simulation pipelines.
- **High-Performance Computing:** Coupled differential equations; numerical integration and finite-difference schemes; HPC workflows (SLURM); sensitivity and convergence analysis.

Referees

Available upon request.